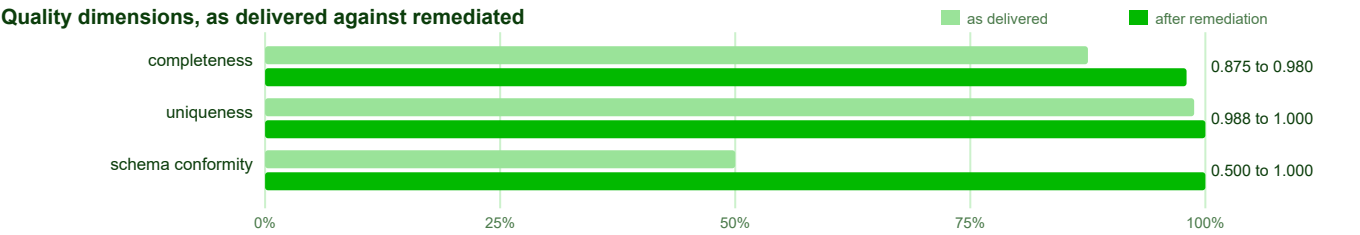


Data Quality Report - spesa.csv

Generated 2026-09-02 20:08 | domain Trattamento_economico | language it

Verdict

Reliability 0.756 to 0.993 over completeness, uniqueness, schema_conformity. The score is a geometric mean, so one broken dimension pulls it down rather than being averaged away. Not measurable on the file as delivered, and therefore outside this figure: validity, consistency.



This is a NoiPA payroll expenditure dataset covering tax and deduction amounts across Italian public administrations. After remediation it is in good, usable shape: structural defects are gone, consistency and uniqueness are fully resolved, and the only remaining weakness is a genuine gap in the area_geografica field. The dataset is clean enough for analysis, with the geographic coverage gap being the one issue that still needs attention at the source.

The dataset as received

measure	value
rows	7,543
columns	18
null cells	16,939
nulls disguised as values	988
duplicate rows	40
rows in key conflict	50
columns badly named	7
columns almost empty	2
columns duplicating another	4
columns still holding the wrong type	0

Completeness read 87.5% on the file as delivered and 86.8% once the placeholders standing in for gaps were counted as gaps. The lower figure is the accurate one.

What was wrong, by coverage area

area	detected	for example
Schema validation	2 names against convention, 5 schema violations	<code>_id</code> , <code>aggregation-time</code>
Completeness	17,337 completeness violations	<code>n.d.</code> , <code>?</code> , <code>//</code>
Consistency	517 cross-column violations, 4 duplicate column groups	<code>ente</code> , <code>tipo_imposta</code> , <code>cod_imposta</code>
Duplicate detection	87 rows not unique when measured, 65 exact duplicates removed	-

area	detected	for example
Anomaly detection	1354 across 2 columns	imposta , spesa
Format validity	513 format violations	descrizione , note , fonte_dato

What the pipeline found

Schema validation

column	suggested name
_id	id
aggregation-time	aggregation_time

column too empty to inform	nulls	null rate
note	7,393	98.0%
fonte_dato	7,468	99.0%

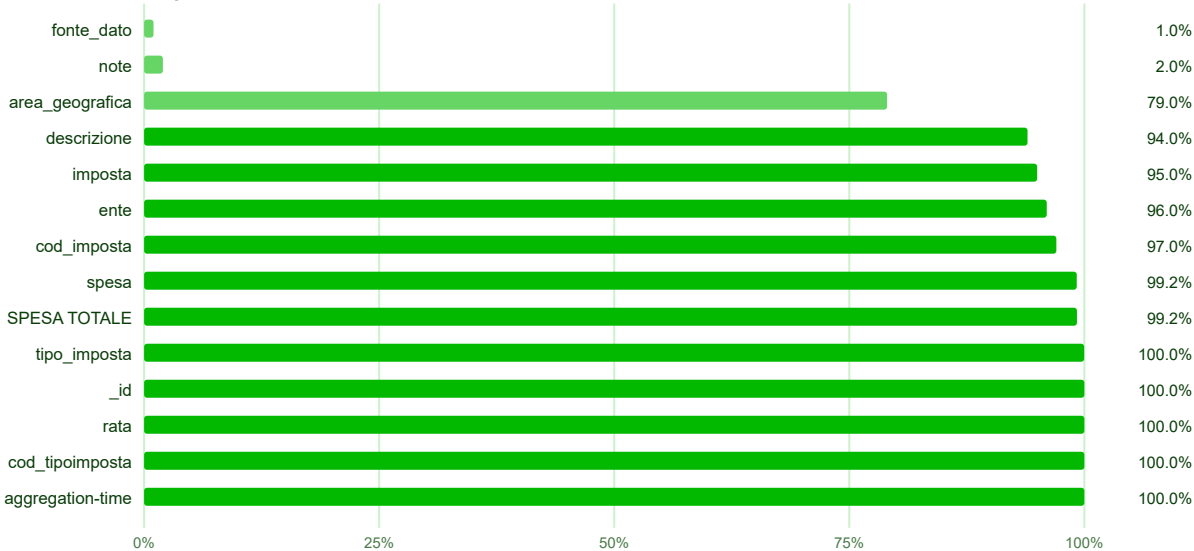
The raw file carried seven badly named columns, two near-empty columns (note and fonte_dato, at roughly 98% and 99% null), and a redundant tax-code column, all of which have now been renamed, dropped, or consolidated. The duplicate-column groups were the most consequential: tipo_imposta and spesa each existed twice under different names, and the surviving versions disagreed with the dropped ones on hundreds of cells. These naming and duplication problems are worth fixing at the source so each field is exported once, under one canonical name.

Completeness

measure	value
overall fill rate	86.8%
null cells	17,927 of 135,774
rows with no gaps	1
rows carrying a gap	7,542

These figures are measured once the placeholders standing in for gaps have been counted as gaps, so the null count is higher here than in the summary of the file as received, which reports what the file appeared to hold.

Fill rate by column, least complete first



column	values that stood in for a gap
descrizione	n.d. , ? , // , ND , , - , unknown , N/A
imposta	- , // , TBD , , unknown , DA VERIFICARE , ? , ND
spesa	N.D.

The apparent completeness of the raw file was flattered by placeholder text such as n.d., TBD, and DA VERIFICARE that counted as values until unmasked; the lower figure is the accurate one, and the drop is a gain in accuracy rather than a regression. After remediation the only meaningful gap is area_geografica, which is missing for about one row in five and cannot be filled without inventing data. The descrizione and imposta gaps were closed deterministically from ente and cod_imposta lookups, leaving only a handful of unresolved entity names.

Consistency

group kept as	data taken from	columns removed	cells backfilled	cells overwritten	values lost
ente	ente%code	ente	0	0	0
tipo_imposta	Tipo Imposta	tipo_imposta	0	383	5
cod_imposta	2cod_imposta	cod_imposta , cod imposta ext	0	0	0
spesa	SPESA TOTALE	spesa	0	37	10
rule	rows breaking it	still breaking it after remediation			
---	---	---			
aggregation-time determines rata	510	0			
cod_imposta determines imposta	4	0			
imposta determines tipo_imposta	1	0			
imposta determines cod_tipoimposta	1	0			
imposta determines cod_imposta	1	0			

The file as delivered held 40 exact duplicate rows, and 65 were removed. The difference is not a discrepancy: collapsing the duplicate columns left further rows identical to one another that had differed only in the columns that were dropped.

Cross-column rules were almost entirely violated by the rata field, where 510 rows had an installment period inconsistent with the aggregation timestamp; these were corrected from the dependency. Duplicate-column resolution did overwrite cells and lose values: the surviving tipo_imposta column disagreed with the dropped one on 383 cells, and the surviving spesa column overwrote 37 cells while discarding values that existed only in the dropped column, including a clearly erroneous -45000.0 and a suspicious 1234567890.0. Sixty-five exact duplicate rows were removed with no conflicting data among them.

Anomaly detection

column	method	detected	for example
imposta	rare_category	2	imposta x , Altro

column	method	detected	for example
spesa	iqr	1,352	2110811.34 , 43365008.73 , 2192935.66

An outlier is unusual, which is not the same as wrong: these are reported and, unless the value is impossible for the column's meaning, left for a person to judge.

The spesa outliers flagged by the IQR method are mostly plausible high-value payroll transactions rather than errors, since expenditure aggregates legitimately span a wide range; however the extreme values near 43 million and the placeholder-like 1234567890.0 deserve a manual look as possible data-entry inflation. In imposta, the rare categories "imposta x" and "Altro" are free-text or placeholder entries that were correctly mapped back to their standard tax labels via the cod_imposta lookup.

Remediation

measure	value
corrections applied automatically	5
proposals put to the reviewer	6
proposals accepted	6
proposals carrying a generated function	0
generated functions validated in a sandbox	0
cells changed in total	5,659
issues carried without an action	1

Issues carried without a corrective action

These were detected and reported. No correction is proposed for them, because none can be expressed as code over the columns the file actually contains; acting anyway would mean inventing values. They are listed so the gap is visible rather than silently carried.

column (rows affected)	why no action is proposed
cod_tipoimposta (1), area_geografica (1,582), tipo_imposta (384), spesa (59), cod_imposta (1)	The missing values in area_geografica (1582), note (7393), fonte_dato (7468), and spesa (59) cannot be safely filled: none of these columns has an imputation hint, and filling them would require inventing data or applying a statistic without a deterministic rule. area_geografica, note, and fonte_dato have no predictor columns in the group, and spesa is a monetary amount that must not be imputed without an explicit user-provided rule. These require human judgement. (aggregation-time, note, fonte_dato are also named above but are covered by a proposal at the gate.)

The pipeline repaired installment-period formatting, rounded floating-point noise in spesa, and imputed descrizione and imposta from high-purity lookups, while leaving area_geografica and the residual spesa gaps untouched because no deterministic rule exists to fill them. No cleaning function was generated for future use, so the repairs are one-off corrections rather than reusable rules. The two near-empty columns were dropped as a schema decision, which is appropriate given they carried almost no information.

What was changed

Applied without asking, because the data determined them

column	correction	cells	why it needed no approval
rata	normalize_period	414	alternative period layouts rewritten to the canonical YYYYMM form

column	correction	cells	why it needed no approval
spesa	round_decimals	2,878	the column is recorded at 2 decimals; the extra digits are floating-point noise and rounding leaves the totals unchanged
rata	complete_period_from_dependency	96	values naming only a year were completed from a column that determines rata exactly
descrizione	impute_from_lookup	448	ente -> descrizione: purity=0.99, coverage=0.98 on path 'raw'
imposta	impute_from_lookup	379	cod_imposta -> imposta: purity=1.00, coverage=1.00 on path 'raw'

Put to the reviewer

id	columns	what it does	outcome
schema_drop_note	note	Drop 'note': it is 98.0% null and carries almost no information.	accepted
schema_drop_fonte_dato	fonte_dato	Drop 'fonte_dato': it is 99.0% null and carries almost no information.	accepted
schema_rename__id	_id	Rename '_id' to 'id' to match the naming convention.	accepted
schema_rename_aggregation-time	aggregation-time	Rename 'aggregation-time' to 'aggregation_time' to match the naming convention.	accepted
impute_from_lookup_descrizione	descrizione	Fill missing descrizione values from the ente-to-descrizione mapping derived from non-missing rows of the dataframe.	accepted
replace_values_imposta	imposta	Correct the placeholder 'imposta x' and the stray 'Altro' value in imposta using the cod_imposta lookup.	accepted

Cells changed, by column

column	cells changed
spesa	2,979
descrizione	758
imposta	625
rata	510
tipo_imposta	387
ente	185
cod_imposta	154
SPESA TOTALE	60
aggregation-time	1

The dataset as delivered

measure	as received	after remediation
rows	7,543	7,478
columns	18	11
null cells	16,939	1,633
format violations	n/a	0
inconsistent rows	n/a	0

measure	as received	after remediation
duplicate rows	40	0
rows in key conflict	50	0
columns badly named	7	0
columns almost empty	2	0
columns duplicating another	1	0
columns still holding the wrong type	0	0

Restricted to the columns present at both ends, reliability moved from **0.938** to **0.996**. The headline pair counts the columns the pipeline removed; this one does not, so removing an empty column is not read as an improvement.

Still open: 1633 completeness.

Every column at a glance

One row per column of the delivered file. `detected` counts what was found against it on arrival, `outstanding` what a check still reports, and `cells changed` how many of its values the run rewrote.

column	type	filled	detected	outstanding	cells changed
<code>id</code> from <code>_id</code>	object	100.0%	0	0	0
<code>rata</code>	object	100.0%	1,020	0	510
<code>descrizione</code>	string	99.9%	456	8	758
<code>cod_tipoimposta</code>	int64	100.0%	1	0	0
<code>imposta</code>	object	100.0%	386	0	625
<code>aggregation_time</code> from <code>aggregation-time</code>	datetime64[ns]	100.0%	0	0	0
<code>area_geografica</code>	object	79.0%	1,582	1,567	0
<code>tipo_imposta</code>	object	100.0%	384	0	387
<code>spesa</code>	float64	99.2%	59	58	2,979
<code>cod_imposta</code>	int64	100.0%	1	0	154
<code>ente</code>	int64	100.0%	0	0	185

Recommendations

- Populate `area_geografica` at the source: require the macro-area (Nord, Centro, Sud, Isole) to be derived from the entity's registered address at data-entry time, since it is currently missing for roughly one in five rows and cannot be safely imputed downstream.
- Enforce a single canonical export for each field: `tipo_imposta` and `spesa` are each emitted twice under different names with conflicting values, so the source system should write one column per concept using the agreed naming convention.
- Validate `spesa` against a plausible range at entry, rejecting negative amounts and flagging values above a domain-specific ceiling, since the data contained a -45000.0 and a 1234567890.0 placeholder.
- Replace free-text tax labels in `imposta` with the `cod_imposta`-driven enum so values like 'imposta x' and 'Altro' cannot be entered, keeping the label fully determined by the code.